MIL-F-81254(WP)
15 April 1965

SUPERSEDING
WS 1703A
6 November 1962

MILITARY SPECIFICATION
FOAM, URETHANE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one type of inert-gas-expanded, isocyanate urethane foam.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated,

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the issue in effect on date of invitation for bids or request for proposal shall apply.

PUBLICATIONS

American Society for Testing and Materials

ASTM D 1622-63	Test for Apparent Density of Rigid Cellular Plastics
ASTM D 570-63	Test for Water Adsorption of Plastics
ASTM D 1621-59T	Tentative Method of Test For Compressive Strength of Rigid Cellular Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of foam system components and foamed material shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials and processing of materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.3 Material. The material shall be a rigid, isocyanate type, closed-cell urethane foam, and shall be the product of a two-component, self-curing foam system.

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3.4 Physical and chemical properties. The physical and chemical property values of the material shall be as specified in Table I.

Table I - Physical and Chemical Properties

Property	Values	
	Minimum	Maximum
Density, pounds per cubic foot	3.14	4.52
Water adsorption, pounds per square foot of surface area per day	--	0.125
Exotherm, degrees Fahrenheit (F)	--	180
Compressive strength, pounds per square inch (psi)	75	106

3.4 Workmanship. The foam system components shall be uniform in quality and shall be free from impurities and other defects that could adversely affect their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the material shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

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4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of sufficient quantity of urethane foam and of foam system components manufactured in accordance with 3.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the foam system by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the tests as specified in 4.5.

4.3.3 Inspection lot. Inspection lots of the foam system components shall consist of all material of each component presented for acceptance at one time, each produced in a single manufacturing batch under homogeneous conditions of manufacture.

4.4 Preproduction inspection. The preproduction sample, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests.

4.4.1 Preparation of test specimens. A foamed material block shall be fabricated in accordance with the manufacturer's instructions, except that the mixing and foaming operations shall be performed at 75 ± 5 degrees F using raw materials preconditioned to 75 ± 5 degrees F. A 5- by 7-3/8 by 5-inch mold shall be used to produce the material sample. The block of foamed material shall be cut into 2-inch cubes for use as test specimens. Each test specimen shall be skinless, and the grain direction which is perpendicular to the free-foamed surface of the original block shall be marked on each test specimen. No less than 5 test specimens shall be used for each test.

4.4.2 Exotherm. Five individual tests shall be performed and no individual value shall exceed 180 degrees F. Temperature indications shall be calibrated within 4 degrees F. An aluminum test block (6061,T6) having the dimensions 4 inches by 3-1/2 inches by 2 inches deep with a 2-inch diameter by 1-1/4-inch deep hole located in the center top surface of the

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block must be used. Position the temperature sensing end of the temperature indicator halfway down the centerline of the hole. Each test shall be performed by mixing 35 grams of the foamed material in accordance with the manufacturer's instructions and pouring into the hole in the aluminum block. Precondition the raw materials and aluminum block at 75 ± 5 degrees F and perform the mixing and foaming operation at this temperature.

4.5 Quality conformance inspection

4.5.1 Visual examination. Visually examine all containers and their contents in the lot to determine compliance with 3.4 and Section 5.

4.5.2 Test specimens. Preparation of test specimens for quality conformance inspections shall be as specified in 4.4.1.

4.5.3 Density test. The test for density shall be performed in accordance with ASTM D 1622-63.

4.5.4 Compressive strength test. The test for compressive strength shall be determined in accordance with ASTM D 1621-59T, procedure A.

4.5.5 Water adsorption test. The test for water adsorption shall be performed in accordance with ASTM D 570-63, procedure (a), using test specimens prepared as specified in 4.4.1. Specimen surface area shall be calculated from the dimensions of the specimen.

4.6 Acceptance criteria

4.6.1 Preproduction. Failure of any sample to comply with any requirement of this specification shall be cause for rejection of the sample.

4.6.2 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified in the contract or order, minimum requirements for preservation, packaging and packing shall be in accordance with MIL-P-116, method III.

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5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as a component of rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Whether preproduction sample is required
d. Place of delivery
e. Size of container
f. Request for test data
g. Request for duplicate copies of detailed instruction sheets for foaming-in-place including data on mold preparation, mixing and foaming procedures, time for self-curing, mix ratio, set time and foam time.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-F-21254(MR) FOAM, URETHANE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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